Jian Yong Zheng Wu

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Education

Instituto Tecnológico de Costa Rica — Costa Rica
Bachelor's Degree in Computer Engineering — *2023 – 2028 (in progress)*

New Way Highschool — Costa Rica
High School Diploma — *2017 – 2022*

Experience

Laboratory Assistant — 07/2023 – 12/2023
Laboratorio de Investigación LuTec, Instituto Tecnológico de Costa Rica

- Worked under the supervision of Milton Villegas Lemus, Computer Engineering Program Coordinator
- Assisted in research and experiments related to Computer Engineering
- Collaborated with professors and researchers on engineering projects

Projects

Custom Compiler and RISC Processor System — 2026

- Co-designed a programming language and built its compiler from scratch, including lexical, syntactic, and semantic analysis, intermediate representation, and custom RISC assembly generation
- Implemented loop unrolling, liveness-based dead code elimination, register renaming, and dependency-aware instruction scheduling
- Contributed to integration with a five-stage RISC pipeline and L1/L2 cache hierarchy, evaluating compiler correctness and optimization impact through code size, instruction counts, compilation time, and simulated execution cycles

Heterogeneous Real-Time Scheduling System — 2026

- Developed a Linux-hosted system with heterogeneous microcontrollers running timer-driven RMS and EDF schedulers for periodic and sporadic tasks
- Implemented node calibration, First-Fit/Worst-Fit task allocation, schedulability checks, and deadline monitoring
- Integrated CRC-validated serial communication and host process synchronization using POSIX shared memory and semaphores

F1 Garage Manager (Full-Stack System) — 2026

- Built a Formula 1 management system using React and Node.js, with role-based access and race simulation
- Designed a normalized SQL database with stored procedures and transactions, and integrated Grafana dashboards for data visualization

Certifications

MATLAB Onramp – MathWorks (2025)

Circuit Simulation Onramp – MathWorks (2025)

Technical Skills

Languages: C, C++, Python, Java, JavaScript, C#, Assembly (ARM, RISC-V), SystemVerilog, MATLAB, Octave, Racket, Prolog
Web & Databases: React, Node.js, HTML, CSS, SQL, MongoDB, Relational Database Design
Architecture: ARM and RISC-V, FPGA Development, Digital Systems Design
Tools & Testing: Git, GitHub, Docker, CMake, GoogleTest, GDB, QEMU, Quartus, ModelSim, Grafana, Jupyter, Jira, Azure DevOps
AI-Assisted Development: Use of AI tools to support coding, debugging, documentation, and exploration of technical solutions, with critical review and validation of generated outputs
Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Agile Methodologies

Languages

English: Advanced

Spanish: Native

Mandarin/Cantonese: Intermediate